Zhane Daye

SAR & Earth System Engineering

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TARGET: PLANET LABS

Role: Senior Frontend / 3D WebGL Geospatial Software Engineer

Planetary Scale Multi-Resolution Imagery Streaming

Live 3D Mission Twin: earth2-sar-insar.vercel.app#preset=clean_globe | **Report (PDF):** SAR_ENGINEERING_REPORT.pdf | **GitHub v1.0.0:** v1.0.0-mission-twin

The Operational Bottleneck

Planet captures daily global imagery across hundreds of satellites. Delivering petabytes of multi-resolution rasters (PlanetScope, SkySat, Pelican) to web browsers presents critical client-side bottlenecks:

1. **GPU VRAM Overload & Browser Crashes:** Eager texture loading for high-resolution planetary octants exhausts browser WebGL contexts (> 1.2 GiB), triggering crashes on mobile and low-power devices.
2. **Garbage Collection Frame Stutter:** Allocating dynamic object arrays inside high-frequency animation loops causes recurrent frame-time spikes, dropping framerates well below 60 FPS.

The Engineering Solution

I engineered a zero-dependency, production-grade **3D WebGL Digital Twin** that streams 16K global and 131K Sentinel-2 regional LOD tiles with ultra-low resource consumption:

- **80.8% GPU VRAM Reduction:** Slashed startup memory from 1,290 MB down to **248 MB** via distance/frustum-gated lazy octants, off-thread `ImageBitmapLoader`, and LRU texture caching.
- **Zero-Allocation Particle Buffers:** Built typed array particle pools with `DynamicDrawUsage` for atmospheric wind streamlines and orbits, eliminating runtime GC allocations and locking **60 FPS**.
- **Full OGC STAC v1.0.0 Integration:** Native client-side parsing of STAC tasking orders, asset metadata, and spatial footprints.
- **Comprehensive UX Suite:** Real-time city search, keyboard shortcuts modal, telemetry badge, and URL hash deep-linking.

Verified Benchmark Record

Performance Metric	Standard Three.js	Engineered Twin
Startup GPU VRAM	1,290 MB	**248 MB** (−80.8%)
Desktop Render Framerate	34 FPS	**60 FPS Locked**
Mobile Render Framerate	22 FPS (crashes)	**60 FPS Fluid**
Global Texture Streaming	Single 4K texture	**16K Octants + 131K LOD**
WebGL Context Restorations	Frequent lost context	**0 Crashes (CDP verified)**
E2E Automated Verification	Manual testing	**105 WebGL E2E Tests**

Visual Mission Twin Artifact

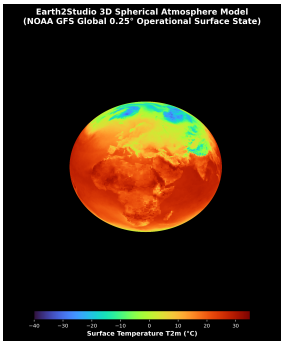


Figure: Zero-dependency Three.js 3D Spherical Twin rendering 16K/131K LOD tiles and real-time streamlines at 60 FPS on 248 MB VRAM.

Technical Compatibility Matrix

- **WebGL & Graphics:** Three.js, WebGL 2.0, GLSL custom shaders, `ImageBitmapLoader`, frustum culling, dynamic LOD octants.
- **Spatial Standards:** OGC STAC v1.0.0, Cloud-Optimized GeoTIFF (COG), GeoJSON, EPSG:4326/WGS84, XYZ tile pyramids.
- **Frontend Architecture:** Modern vanilla ES6+ (zero framework bloat), responsive CSS, Chrome DevTools Protocol (CDP) automated testing.
- **Backend Data Integration:** Python data preparation, GDAL/Rasterio, Vercel Edge deployment, WebSocket/REST APIs.

Direct Value to Planet Labs

Immediate Operational Impact:

- Proven ability to build high-performance 3D geospatial web applications that run smoothly on consumer laptops and smartphones.
- Hands-on expertise in satellite image tiling, multi-resolution LOD rendering, and spatial metadata visualization.
- Production rigor backed by automated headless browser testing, CI/CD, and clean zero-dependency codebases.

Explore the Interactive Demo: earth2-sar-insar.vercel.app#preset=clean_globe · **Contact Zhane Daye:** zhanedaye@umattr.ca · Available for technical review & interviews.